

Pollinator Garden Suitability analysis for Auraria Campus



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Introduction

Pollinators such as bees and butterflies play an important role in supporting biodiversity and maintaining healthy ecosystems. However, urbanization and habitat loss have contributed to declining pollinator populations in many cities. As urban campuses continue to expand, incorporating pollinator-friendly green spaces can help improve habitat availability, and promote more sustainable urban environments.

Auraria Campus contains a variety of landscaped and open areas that may be suitable for a pollinator garden. Identifying the most appropriate locations requires consideration of several environmental and anthropogenic factors, including sunlight availability, pedestrian traffic and available landscape space.

A GIS-based suitability analysis was conducted to identify potential locations for a pollinator garden across Auraria Campus. Areas were evaluated based on the selected criteria to identify locations with the greatest potential for a successful pollinator habitat.

Background

Pollinators are responsible for supporting many flowering plants and agricultural crops, making them essential to both ecosystems and food production. Despite their importance, pollinator populations have declined due to habitat fragmentation, urbanization, pesticide use, and loss of native vegetation. Urban Environments often contain limited habitat and food sources for pollinator species, creating challenges for long-term population sustainability.

Pollinator gardens are increasingly being incorporated into urban sustainability initiatives because they can provide habitat, food resources, and ecological connectivity for pollinator species. College Campuses present unique opportunities for pollinator conservation due to the presence of landscaped green spaces and opportunity for community engagement and education

Methods

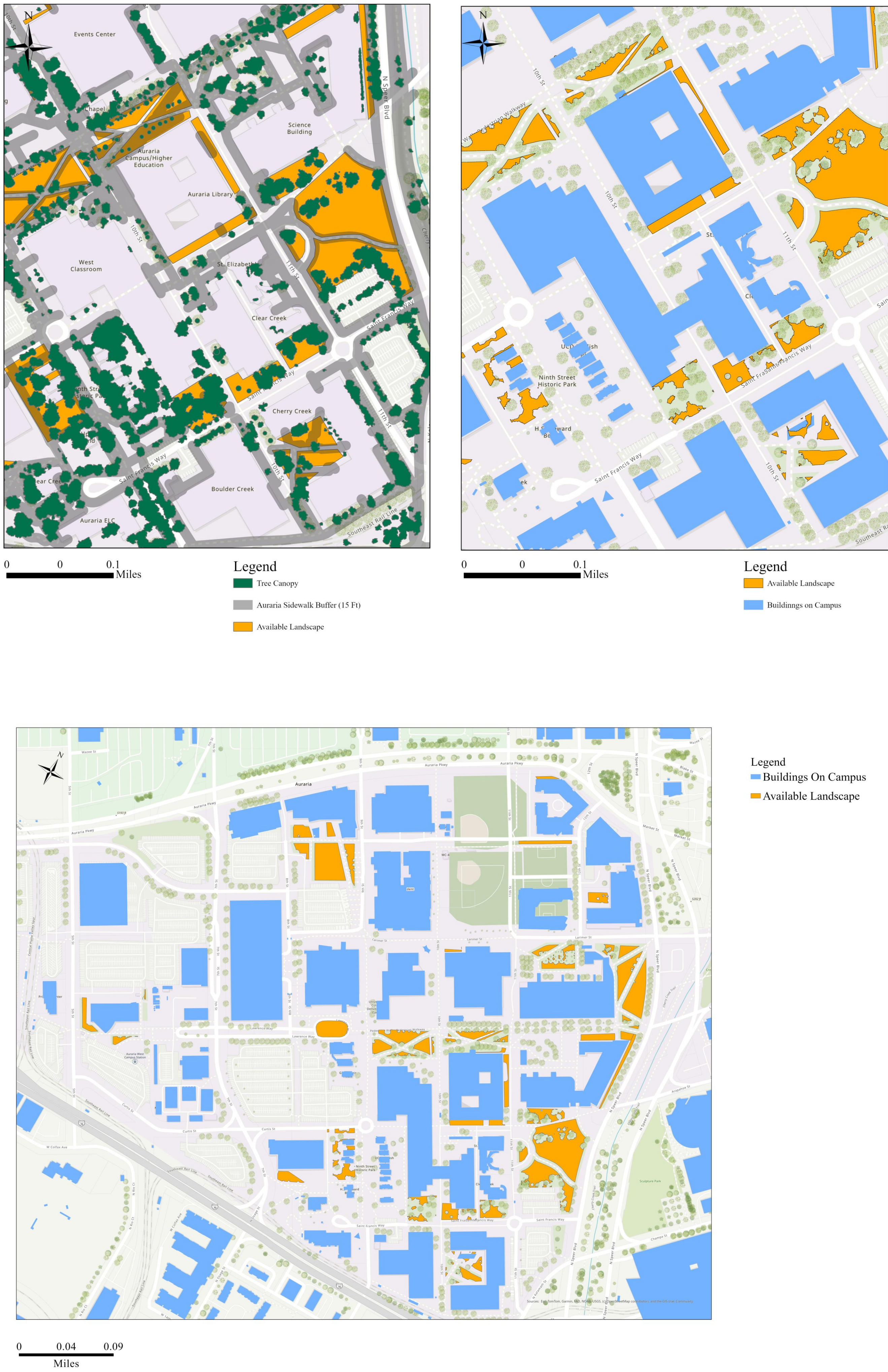
Using ArcGIS Pro a suitability analysis was conducted to identify potential locations for a pollinator garden on Auraria Campus. Spatial data related to tree canopy and campus walking trails were obtained from the City and County of Denver Open Data Catalog processed for analysis. Information regarding landscaped and open areas on campus was provided by campus staff, and polygon features representing these areas were manually digitized for use in the analysis.

A 15-foot buffer was created around sidewalks and walking trails to represent areas of high pedestrian traffic and disturbance. Areas farther away from heavily trafficked walkways were considered more suitable for pollinator habitat. Tree canopy data was used to estimate sunlight availability, with areas containing less canopy were considered more suitable for pollinator-friendly vegetation.

The selected criteria were combined using suitability analysis techniques to identify locations across Auraria Campus with the highest potential for a successful pollinator habitat.

Results

The suitability analysis identified several potential locations for a pollinator garden at Auraria Campus. Suitable areas were primarily concentrated around existing landscaped spaces and along building perimeters throughout campus. Several larger connected area were identified near central campus green spaces, while smaller suitable patches were distributed throughout the study area.



Discussion

The analysis identified several areas across Auraria Campus that may support a future pollinator garden. The resulting maps demonstrate how GIS can be used to support campus sustainability planning and identifying potential opportunities for urban pollinator conservation.

While the analysis produced several suitable locations, additional environmental factors and field verification could improve the accuracy of the project. Future research could incorporate variable such as soil conditions, irrigation access, native plant selection, pesticide use, and seasonal environmental changes.

Conclusion

This project identified potential locations for a pollinator garden on Auraria Campus using GIS and spatial analysis techniques.

By evaluating factors related to available landscape space, pedestrian disturbance, and tree canopy cover, several areas across campus were identified as potentially suitable for a pollinator habitat.

The results of this study may support campus sustainability and pollinator conservation initiative while also providing guidance for pollinator garden planning on campus.

Sources

Building outlines 2022. Open Data Catalog. (n.d.-a). https://opendata-geospatialdenver.hub.arcgis.com/datasets/04e5069b4c843d2bfc15a0d1c66e00_111/explore

Klein, A.-M., Cunningham, S. A., Bos, M., & Steffan-Dewenter, I. (2008). Advances in pollination ecology from tropical plantation crops. *Ecology*, 89(4), 935–943. <https://doi.org/10.1890/07-0088.1>

Penn, H., Penn, J., Hagan, M., & Hu, W. (2020). The buzz about Bee campuses: Student thoughts regarding pollinator conservation. *American Entomologist*, 66(4), 54–61. <https://doi.org/10.1093/ae/tmaa055>

Sidewalk line 2020. Open Data Catalog. (n.d.-a). https://opendata-geospatialdenver.hub.arcgis.com/datasets/7c2101ad2822408ca4339ce16a49a2e9_333/explore?location=-39.731208%2C-104.888846%2C10

66(4), 54–61. *Statistical neighborhood*. Open Data Catalog. (n.d.). https://opendata-geospatialdenver.hub.arcgis.com/datasets/3e57d472a8b4326867c1a4c9447c91_13/explore?location=-39.763943%2C-104.854695%2C10

Tree Canopy 2020. Open Data Catalog. (n.d.-c). https://opendata-geospatialdenver.hub.arcgis.com/datasets/40e936c38214862ad0d129039733dc5_348/explore?location=-39.763949%2C-104.855657%2C10